Behavioral Emotional Interference Changes Following Gender-Affirming Hormone Therapy Initiation

(Individualized Bayesian Region-of-Practical-Equivalence Approach)

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Master of Science Thesis Major Research (BS030)

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Abstract

Starting gender-affirming hormone therapy (GAHT) is associated with affective and psychosocial changes, but it remains unclear whether early GAHT is accompanied by changes in behavioral responses to emotionally conflicting social cues. This longitudinal observational thesis assessed 13 trans and gender-diverse adults on the day of GAHT initiation and again three months later. General emotional interference was defined as slower reaction times on incongruent than congruent angry–happy face-word Emotional Stroop trials. Participant-level Bayesian mixed-effects models were estimated and then synthesized within masculinizing and feminizing GAHT groups using a Bayesian Split/Analyze/Meta-Analyze approach. Effects significance was interpreted using a preregistered region of practical equivalence of ±0.1 standardized effect-size units. Two participants showed practically significant general emotional interference, and aggregate estimates suggested a small positive interference effect, strongest in the masculinizing group. Valence-specific estimates were heterogeneous: two masculinizing-GAHT participants showed happy-distractor-biased interference, which pulled the group estimate toward a negative trend. Evidence for three-month change in either general interference or valence-specific bias was inconclusive, and exploratory associations with depressive symptoms were not robust. These findings suggest that early GAHT was not accompanied by a systematic change in emotional interference in this small heterogeneous sample, while highlighting person-level variability and importance of individual modelling.
Keywords: gender-affirming hormone therapy, transgender and gender-diverse adults, emotional interference, face-word emotional Stroop, Bayesian mixed-effects modelling

Introduction

Some individuals, who (might) identify as transgender and gender-diverse (TGD)[1], experience persistent incompatibility between their self-perceived gender identity and bodily characteristics, clinically termed gender incongruence (Coleman et al., 2022). Consequently, when desired and accessible (Berrian et al., 2025), some TGD people initiate and sustain feminizing or masculinizing gender-affirming hormone therapy (GAHT) to induce changes in bodily features (e.g., body fat, hair, muscle mass) that over time help bring the body into better alignment with the gender identity (Coleman et al., 2022; Hembree et al., 2017; Nguyen et al., 2018a). Recent evidence syntheses also established a strong support for reductions in depressive symptoms and psychological distress following GAHT initiation (Baker et al., 2021; Doyle et al., 2023; Van Leerdam et al., 2023; White Hughto & Reisner, 2016).

Consequently, GAHT is increasingly conceptualized not merely as an exogenous gonadal hormone intervention, but as a broader psychosocial transition potentially connected to substantial shifts in psychological (e.g., wellbeing, self-perception, emotional processing) and interpersonal functioning (i.e., the quality of social relationships, communication, and social support; Doyle et al., 2023; Doyle & Link, 2024; Millet et al., 2026; Morssinkhof et al., 2025a). Yet, the timing, direction, and mechanisms of psychosocial change after GAHT remain underexplored and heterogeneous (Doyle et al., 2023). Moreover, prospective work has not examined to date whether GAHT initiation is accompanied by changes in behavioral responses to social-emotional conflict. The present thesis addressed this gap by assessing emotional interference in TGD adults on the day of clinically supervised GAHT initiation, and again three months later, with the aim of understanding for whom and how social-emotional behavioral patterns change over time.

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[1] Transgender and gender-diverse refers to individuals whose gender identity and/or expression does not align with gendered expectations associated with their sex assigned at birth (Nguyen et al., 2018a). Cisgender individuals are those whose gender identity is congruent with their sex assigned at birth (Nguyen et al., 2018a).

Feminizing or Masculinizing Gender-Affirming Hormone Therapy

Depending on individual gender-affirming goals, GAHT is broadly categorized into masculinizing and feminizing gonadal hormone regimens (Coleman et al., 2022; Hembree et al., 2017). Accordingly, feminizing GAHT (F-GAHT) aims to elevate estradiol and reduce testosterone, and vice versa in Masculinizing GAHT (M-GAHT), as TGD individuals seek to induce bodily characteristics culturally read as more masculine, feminine, androgynous, or nonbinary (Coleman et al., 2022; Hembree et al., 2017). M-GAHT generally involves testosterone administration, which may include agents for menstrual suppression, whereas F-GAHT typically involves estradiol administration, often accompanied by suppression of endogenous testosterone through anti-androgenic medication (Coleman et al., 2022; Hembree et al., 2017; Irwig, 2016; Kuijpers et al., 2021).

Therefore, people initiating GAHT differ in dose, route of administration, adjunctive medication, prior occasionally self-administered endocrine histories, and embodiment goals (Coleman et al., 2022; Mepham et al., 2014; Nguyen et al., 2018a). Moreover, creating testosterone- or estrogen-dominant endocrine milieus may each exert distinct functional effects on brain systems supporting social-emotional information processing (Hornung et al., 2020; Nguyen et al., 2018b). Indeed, gonadal hormones, that is, testosterone and estradiol, have been shown to exert potential modulatory effects on brain structures that subserve emotional processing, regulation, and cognitive control (Barth et al., 2015; Hornung et al., 2020; Kranz et al., 2020; Song et al., 2017).

The Modulatory Influence of Gonadal Hormones on Social-Emotional Behaviors

Prior research establishes that administering gonadal hormones could be indirectly or even directly linked to "higher-level" psychosocial functioning outcomes, such as via behavioral responses to emotionally conflicting social stimuli (Barth et al., 2015; Heany et al., 2015). Moreover, gonadal hormones seem to bias certain aspects of social-emotional behavior, defined by how people attend to, interpret, and eventually respond in the face of emotionally salient social cues (Bos et al., 2011; Heany et al., 2015; Ochsner & Gross, 2005), including, for instance, how quickly one approaches or avoids emotionally salient social stimuli (Enter et al., 2014, 2015). Developmental studies suggest that endogenous gonadal hormone levels relate to emotional-conflict-related neural responses during adolescence, and pubertal testosterone has been linked to shifts in neural systems supporting emotional action control (Cservenka et al., 2015; Hornung et al., 2020; Peper et al., 2011; Peper & Dahl, 2013; Tyborowska et al., 2016, 2024; Volman et al., 2011).

Testosterone- and estradiol-dominant endocrine milieus may also differ in how they bias actions such as approaching or avoiding angry or happy cues (Hornung et al., 2020; Volman et al., 2011). However, much of the direct behavioral evidence linking gonadal hormones to social-emotional behavior seems to originate from acute testosterone studies, with more limited evidence for estradiol. In women, acute testosterone administration has been shown to bias approach-avoidance behavior. In double-blind, placebo-controlled designs, single-dose testosterone reduced behavioral avoidance of angry faces in "healthy" women and alleviated gaze avoidance toward socially threatening facial expressions in women with social anxiety disorder (Enter et al., 2014, 2015). Estradiol effects in men are less studied, but acute estradiol administration in "healthy" men has been linked to heightened vicarious emotional responding, biasing social evaluations more empathetically when observing others in distress (Olsson et al., 2016), suggesting that estradiol could potentially also modulate certain aspects of social-emotional behavior.

However, generalizing from this literature to TGD individuals starting GAHT remains particularly difficult for at least two main reasons (Nguyen et al., 2018b). First, most studies utilize cross-sectional or acute between-subject administration designs, which do not capture within-person adaptation to sustained longitudinal hormonal exposure (Barth et al., 2015; Bos et al., 2011; Heany et al., 2015; Kranz et al., 2020; Peper & Dahl, 2013). Second, the vast majority of gonadal hormone research has been conducted in presumably cisgender samples and moreover often relies on sex-binary cultural frameworks to interpret hormone effects (Bos et al., 2011; Kranz et al., 2020; Massa et al., 2023). Extrapolating from cisgender samples is further complicated by the fact that TGD individuals often possess distinct baseline characteristics before initiating GAHT.

Psychosocially, TGD individuals experience disproportionate discrimination and minority stress, which can shape affective processing through the chronic expectation of hostility (Meyer, 2003; Puckett et al., 2023). Indeed, Kiyar et al. (2022b) suggested that minority stress modulates neural responsivity to emotional stimuli in trans men prior to starting GAHT, indicating that chronic exposure to stigma and social threat may sensitize emotion-processing circuits to negative affective cues. Baseline endocrine histories may also vary in people seeking GAHT due to endocrinological interventions, such as gonadotropin-releasing hormone analogues, to postpone or block puberty (Coleman et al., 2022). Thus, while existing evidence provides a plausible basis for GAHT-related changes in social-emotional behavior, it does not provide directional predictions as to what to expect after starting sustained exposure to gonadal hormones via M- or F-GAHT.

Early Gender Affirming Hormone Therapy and Social-Emotional Experiences

Limited prospective neuroimaging evidence suggests that sustained GAHT can be associated with changes in emotion-processing circuitry. Kiyar et al. (2022a) reported that after 6 to 10 months of M-GAHT, trans men’s neural responses to emotional faces shifted toward a pattern more similar to cisgender men, from a baseline pattern more congruent with sex assigned at birth. However, comparable neuroimaging evidence for F-GAHT remains scarce, and virtually no work has tested whether such neural changes are accompanied by behavioral responses to social-emotional stimuli in either GAHT group (Derntl et al., 2024; Rehbein et al., 2021). Similarly, although Cuéllar-Flores et al. (2024) reported improved processing speed and classic color-word Stroop performance after 12 months of M-GAHT in 15 adolescents, the classic Stroop task does not assess cross-modal emotional conflict, and adolescent findings may not generalize to TGD adults initiating GAHT.

The most recent longitudinal quantitative evidence from TGD cohorts that may approximate the social-emotional consequences of early GAHT comes from prospective work on affect and affect variability[2] (Matthys et al., 2021, Morssinkhof et al., 2025b). Matthys et al. (2021) examined positive and negative affect during GAHT and reported early regimen-unrelated affective changes, with positive affect significantly decreasing at three months for both GAHT groups and negative affect remaining relatively stable to baseline. Complementing these mean-level findings, Morssinkhof et al. (2025b) examined daily affect variability before GAHT and after three and twelve months of treatment. Participants initiating F-GAHT showed increased variability in negative affect, whereas those initiating M-GAHT showed decreased variability at three months, indicating early regimen-specific changes in affective dynamics (Morssinkhof et al., 2025b). By twelve months, however, no differences between GAHT groups remained (Morssinkhof et al., 2025b).

Qualitative evidence adds a complementary first-person perspective on why this early period may matter. People starting GAHT describe changes in emotional intensity, emotion regulation, self-perception, authenticity, agency, and interpersonal relating, including changes they may not have fully anticipated before treatment (Fowler et al., 2023; Millet et al., 2026). Thus, early GAHT appears to involve both measurable affective changes and lived emotional

[2] Affect refers here to momentary positive or negative emotional experience, whereas affect variability refers to the extent to which these affective states fluctuate over time (Jones et al., 2023).

or interpersonal reorientation. What remains unknown is whether these self-reported and neural changes are reflected in behavioral performance when people must resolve emotionally conflicting social cues.

Emotional Interference and Valence-Specific Distractor Bias

To test this behavioral gap, the present thesis examined emotional interference in a face-word emotional Stroop paradigm, where social-emotional conflict is operationalized as the reaction time (RT) cost of responding to emotionally incongruent compared with congruent face-word information (Agustí et al., 2017; Etkin et al., 2006; MacLeod, 1991; Smolker et al., 2021; Song et al., 2017). Emotional interference cannot by itself explain psychosocial change, but it is an appropriate task-based index for the present question because it tests whether emotionally conflicting social cues disrupt performance during early GAHT, thereby complementing prior self-report evidence on affective and psychosocial change after GAHT initiation (Doyle et al., 2023; Inzlicht et al., 2015; Matthys et al., 2021; Morssinkhof et al., 2025b; Song et al., 2017). This cost is also theoretically relevant to GAHT initiation because face-word emotional Stroop paradigms seem to engage emotional-conflict monitoring and control systems that partly overlap with gonadal-hormone-sensitive circuits involved in social-emotional processing and action control (Barth et al., 2015; Chechko et al., 2009; Etkin et al., 2006; Song et al., 2017; Tyborowska et al., 2016, 2024; Volman et al., 2011).

Accordingly, this thesis is the first prospective study to examine whether initiating GAHT is accompanied by general emotional interference. Meta-analytic results in Supplementary Figure S1 indicated a moderate-to-large (Cohen, 1988) general emotional-interference effect across prior face-word Stroop studies, supporting the expectation that this task can detect a measurable incongruency cost. Thus, research question 1 (RQ1) asked whether participants showed general emotional interference, with the directional hypothesis 1

(H1) predicting slower responding to emotionally incongruent than congruent information, irrespective of whether the distractor was angry or happy (Agustí et al., 2017; Başgöze et al., 2015; Chechko et al., 2012; Fan et al., 2016; Smolker et al., 2021; Song et al., 2017).

Building on RQ1, research question 2 (RQ2) asked whether the general interference cost was valence-neutral or biased toward positively versus negatively valenced distractors (Agustí et al., 2017; Koizumi et al., 2007; Smolker et al., 2021). Valence-specific distractor bias refers here to the difference between angry- and happy-distractor interference, where stronger angry-distractor interference could suggest preferential distractibility by threat-related cues, whereas stronger happy-distractor interference might indicate preferential distractibility by affiliative or approach-related cues (Agustí et al., 2017; Fan et al., 2016; Smolker et al., 2021; Song et al., 2017). This distinction is particularly relevant in TGD adults initiating GAHT because pre-treatment affective processing may already be shaped by minority stress, vigilance, and chronic expectations of hostility, while early GAHT may also alter the salience of social-emotional cues (Fox et al., 2000; Kiyar et al., 2022b; Lukito et al., 2023; Meyer, 2003; Puckett et al., 2023). Because Supplementary Figure S2 showed heterogeneous valence-specific evidence where this contrast was extractable, hypothesis 2 (H2) was specified non-directionally (Agustí et al., 2017; Başgöze et al., 2015; Beall & Herbert, 2008; Chechko et al., 2012; Cothran et al., 2012; Hampson et al., 2024; Koizumi et al., 2007; Kuehne et al., 2019; Ros et al., 2021).

Given that the first three months of GAHT may involve early gonadal-hormone transition, affective change, and psychosocial adjustment, research question 3 (RQ3) extended RQ1 longitudinally by testing whether general emotional interference changed from baseline to three months (Matthys et al., 2021; Millet et al., 2026; Morssinkhof et al., 2025b; Verbeek et al., 2024). Specifically, RQ3 asked whether the overall incongruency cost increased, decreased, or remained stable after GAHT initiation. Hypothesis 3 (H3) was non-directional because existing GAHT evidence indicates early affective and psychosocial change, but does not establish whether behavioral costs in resolving emotionally conflicting social cues should increase, decrease, or remain unchanged during the first three months of treatment (Doyle et al., 2023; Matthys et al., 2021; Morssinkhof et al., 2025b; Nguyen et al., 2018b).

Research question 4 (RQ4) then extended RQ2 longitudinally by testing whether valence-specific distractor bias shifted from baseline to three months. This question was motivated by the possibility that angry versus happy distractor processing may already be altered before GAHT through minority stress and chronic social threat exposure, while sustained GAHT may also alter neural responses to emotional faces over time, as suggested by pre-treatment minority-stress findings and longer-term M-GAHT neuroimaging evidence (Kiyar et al., 2022a, 2022b). RQ4 therefore asked whether the relative angry-versus-happy distractor cost shifted over time toward stronger threat-related or affiliative distractor interference. Hypothesis 4 (H4) was non-directional because no prior prospective GAHT study has established whether valence-specific behavioral interference should shift toward angry distractors, shift toward happy distractors, or remain stable in adults initiating GAHT (Derntl et al., 2024; Kiyar et al., 2022a; Nguyen et al., 2018b; Rehbein et al., 2021).

Potential Clinical Relevance of Face-Word Emotional Stroop Task

Emotional interference may also be clinically informative because emotional Stroop interference has been meta-analytically linked to depression and related psychopathology, although effects vary by task variant, stimulus type, and clinical group (Başgöze et al., 2015; Epp et al., 2012; Joyal et al., 2019). In particular, depressive symptom severity has been associated with stronger behavioral interference effects in emotional Stroop work (Epp et al., 2012), while depressive symptoms are also among the psychosocial outcomes most consistently reported to improve after GAHT (Doyle et al., 2023; Van Leerdam et al., 2023).

Because depression-related Stroop effects have not been established in TGD adults initiating GAHT, the present thesis therefore examined these links exploratorily (Başgöze et al., 2015; Epp et al., 2012; Joyal et al., 2019; Nguyen et al., 2018b).

Specifically, research question 5a (RQ5a) tested whether overall depressive symptom severity was associated with general emotional interference across the baseline-to-three-month window, asking whether participants with higher Center for Epidemiologic Studies Depression Scale (CES-D; Radloff, 1977) scores showed stronger or weaker interference. Research question 5b (RQ5b) tested whether within-person change in CES-D scores co-occurred with within-person change in general emotional interference during the first three months of GAHT. No directional hypotheses were specified for RQ5a or RQ5b because no prospective GAHT study currently establishes how depressive symptom severity or within-person depressive symptom change should map onto behavioral emotional interference in adults initiating GAHT.

Methods

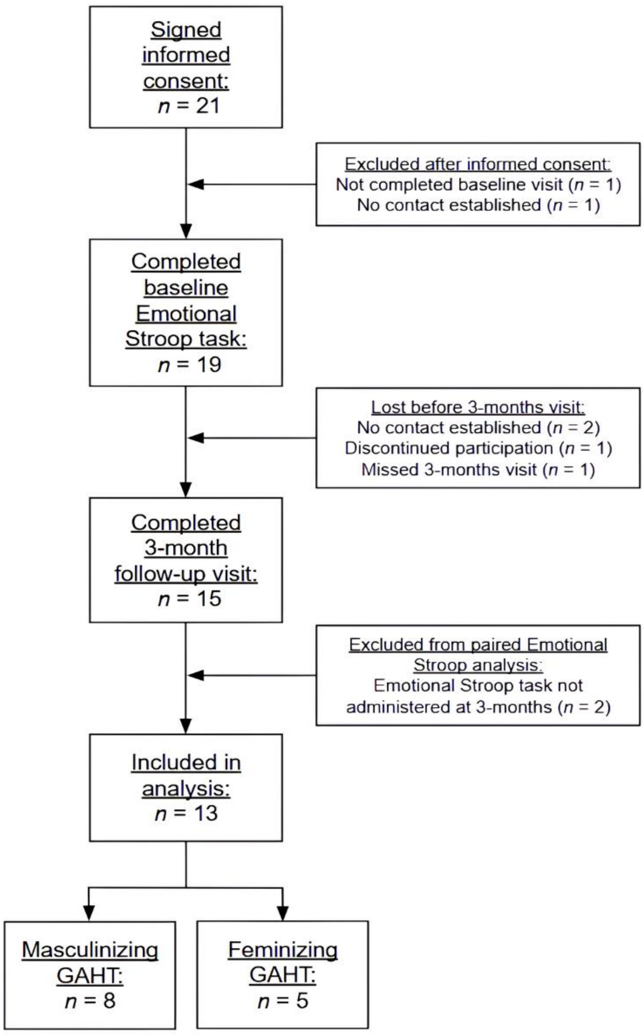
Participants

The analyzed sample consisted of 13 TGD adults who completed both baseline and three-month follow-up questionnaire and in-person study visits, including eight participants initiating M-GAHT and five initiating F-GAHT. Participant flow from informed consent to inclusion in the final analyzed sample is shown in Figure 1. Sociodemographic and clinical characteristics of the analyzed sample, stratified by hormonal treatment group and assessment timepoint, are presented in Table 1. Participants were recruited from the Center of Expertise on Gender Dysphoria at Amsterdam University Medical Center (UMC) in the Netherlands, as part of the larger prospective AFFIRM Relationships (https://www.affirmrelationships.com/) cohort study with published protocol by Morssinkhof et al. (2025a).

Inclusion criteria were being at least 18 years of age, initiating M-GAHT or F-GAHT (for the first time), and having sufficient Dutch or English proficiency to complete the study procedures. Exclusion criteria were intellectual disability or severe psychological comorbidities that, in the judgement of a treating healthcare professional, would render participation excessively burdensome. Participants received monetary compensation in accordance with institutional guidelines (see Data Collection Details in the Supplementary Materials). Ethical approval was granted by the Medical Research Ethics Committee of Amsterdam UMC (study no. 2024.0927).

Figure 1

Participant Flow from Informed Consent to Inclusion in the Final Emotional Stroop Analysis



Note. GAHT = gender-affirming hormone therapy. Twenty-one participants provided informed consent. Two participants were excluded prior to baseline. Nineteen participants completed the baseline Emotional Stroop assessment. Four participants were lost at the 3-month follow-up visit. Fifteen participants completed the 3-month follow-up assessment. Two participants were excluded from the paired Emotional Stroop analysis because the Emotional Stroop task was not administered at the 3-month visit. The final analytic sample comprised 13 participants, 8 receiving masculinizing and 5 receiving feminizing GAHT.

Table 1

Sociodemographic and Clinical Characteristics at Baseline and Three Months Separately for Masculinizing and Feminizing Gender-Affirming Hormone Therapy Groups

Variable	Masculinizing		Feminizing	
	Baseline (*n* = 8)	3 months (*n* = 8)	Baseline (*n* = 5)	3 months (*n* = 5)
Age, years				
Mean (*SD*)	25.63 (10.31)	—	40.20 (17.20)	—
Range (min–max)	18.0–50.0	—	27.0–64.0	—
Gender identity, *n* (%)				
Binary	4 (50.0)	4 (50.0)	5 (100)	5 (100)
Non-binary	4 (50.0)	4 (50.0)	0 (0)	0 (0)
BMI (kg/m²)				
Mean (*SD*)	25.46 (4.34)	28.42 (4.55)	25.50 (2.33)	26.27 (2.83)
Depression (CES-D)				
Mean (*SD*)	13.25 (8.00)	15.00 (9.18)	9.80 (11.97)	14.00 (11.18)
Range (min–max)	5.0–29.0	4.0–28.0	0.0–26.0	5.0–29.0

Autism quotient (AQ10)				
Mean (*SD*)	23.25 (4.56)	23.38 (4.37)	22.60 (5.81)	21.80 (6.98)
Range (min–max)	15.0–29.0	18.0–29.0	16.0–29.0	13.0–30.0
Emotion reactivity (B-ERS)				
Mean (*SD*)	7.25 (4.62)	8.25 (4.53)	7.60 (4.72)	7.20 (5.67)
Range (min–max)	1.0–14.0	4.0–18.0	0.0–12.0	3.0–17.0
Formal autism diagnosis, *n* (%)				
Yes	2 (25.0)	—	2 (40.0)	—
No	6 (75.0)	—	3 (60.0)	—
Self-reported psychiatric diagnosis, *n* (%)				
Any comorbidity	2 (25.0)	5 (62.5)	1 (20.0)	1 (20.0)
Depression or dysthymia	1 (12.5)	1 (12.5)	0 (0.0)	0 (0.0)
Generalized anxiety disorder	0 (0.0)	1 (12.5)	0 (0.0)	0 (0.0)
Social phobia or agoraphobia	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Posttraumatic stress disorder	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Attention-deficit/ hyperactivity condition	0 (0.0)	1 (12.5)	0 (0.0)	0 (0.0)
Autism spectrum condition	2 (25.0)	2 (25.0)	2 (40.0)	2 (40.0)
Self-reported psychotropic medication use, *n* (%)				
Psychotropic medication	1 (12.5)	2 (25.0)	0 (0.0)	0 (0.0)

Antidepressants	1 (12.5)	1 (12.5)	0 (0.0)	0 (0.0)
Antipsychotics	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Anxiolytics	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Stimulants	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Mood stabilizers	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Testosterone form prescribed, *n* (%)				
Testosterone gel	6 (75.0)	3 (37.5)	—	—
Testosterone esters	2 (25.0)	1 (12.5)	—	—
Testosterone undecanoate	0 (0.0)	4 (50.0)	—	—
Cycle regulation, *n* (%)			—	—
Yes	4 (50.0)	4 (50.0)	—	—
No	4 (50.0)	4 (50.0)	—	—
Estrogen form prescribed, *n* (%)				
Estradiol valerate tablets	—	—	3 (60.0)	3 (60.0)
Estradiol gel	—	—	0 (0.0)	0 (0.0)
Estradiol patches	—	—	2 (40.0)	2 (40.0)
Antiandrogen form prescribed, *n* (%)				
Cyproterone acetate	—	—	0 (0.0)	0 (0.0)
GnRH analogs (short-acting)	—	—	0 (0.0)	0 (0.0)
GnRH analogs (long-acting)	—	—	4 (80.0)	4 (80.0)
Spironolactone	—	—	0 (0.0)	0 (0.0)
None	—	—	1 (20.0)	1 (20.0)

Note. GAHT = gender-affirming hormone therapy; *SD* = standard deviation; BMI = body mass index; CES-D = Center for Epidemiologic Studies Depression Scale (Radloff, 1977); AQ10 = Autism Spectrum Quotient (Allison et al., 2012); B-ERS = Brief Emotion Reactivity Scale (Veilleux et al., 2023); GnRH = gonadotropin-releasing hormone. Em dashes (—) indicate data that were not collected or are not applicable for a given GAHT group. Values are restricted to participants who had completed data at both baseline and 3-month follow-up (see Figure 1). Binary gender identities comprise individuals who identified as (trans-)men or (trans-)women; the remaining participants identified as non-binary.

Materials

Face-Word Emotional Stroop Task

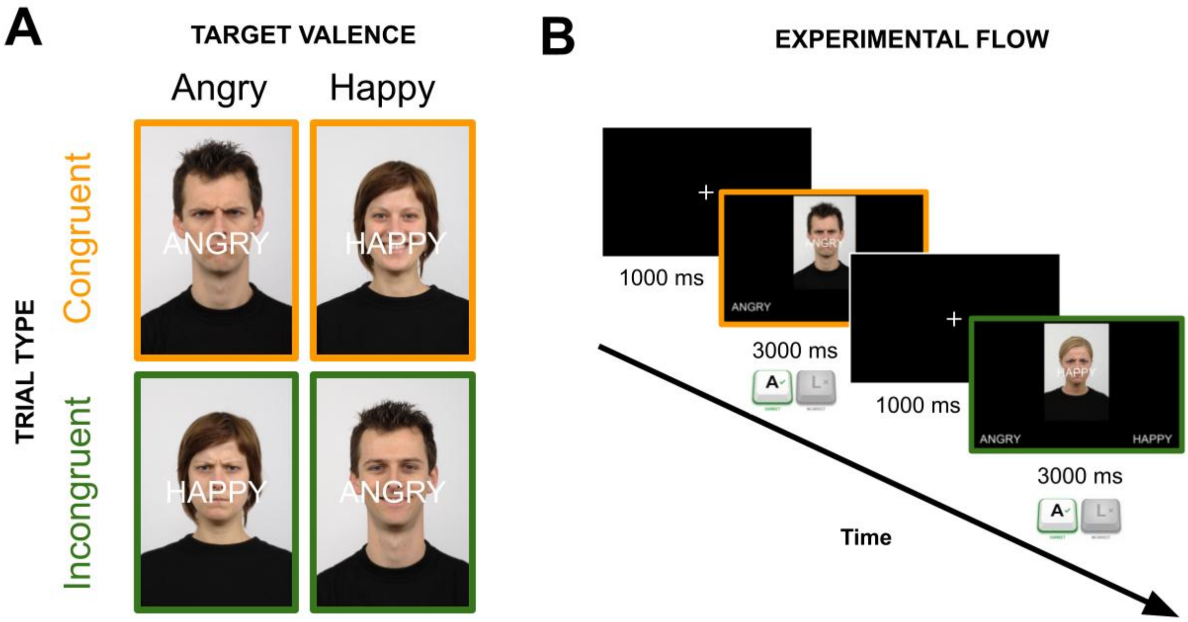
Emotional interference was assessed behaviorally using a computerized implementation of the self-programmed Face–Word Emotional Stroop Task (FWEST; Etkin et al., 2006) via PsychoPy (v2024.2.4; Peirce et al., 2019). The current FWEST parameters were based on the paradigm by Agustí et al. (2017). Facial stimuli consisted of 36 unique identities (18 female, 18 male) selected from the Radboud Faces Database (RaFD; Langner et al., 2010), each displaying happy and angry expressions (i.e., 72 unique faces). On every trial, a facial image was overlaid with a centrally fixed emotion word ("HAPPY" or "ANGRY").

Trials followed a 2 × 2 × 2 factorial design manipulating target emotion (happy vs. angry), model gender (female vs. male), and congruency (congruent vs. incongruent) across trials. A coding error during the early phase of FWEST data collection had minor implications for equal randomization across facial stimuli (see Randomization Error in Supplementary Materials). On congruent trials, the facial expression matched the semantic content of the overlaid word; on incongruent trials, the expression and word mismatched, eliciting cross-modal emotional interference (schematic representation in Figure 2, panel A).

The task was divided into two blocks—a face-target block and a word-target block—each comprising 4 practice and 68 analyzed trials. In the face-target block, participants were instructed to classify the facial expression while ignoring the word, whereas in the word-target block, they classified the word while ignoring the face. Block order was counterbalanced across participants. Responses were recorded via keyboard input during the stimuli presentation (see Figure 2, panel B). The primary dependent variable was RT responses on both accurate and inaccurate trials.

Figure 2

Schematic Overview of Face-Word Emotional Stroop Trial Conditions and Experimental Flow



Note. Panel A illustrates the core congruent–incongruent structure of the Face-Word Emotional Stroop Task, crossing target valence (angry vs. happy) with congruency (congruent vs. incongruent). The example depicts the face-target block, in which participants classify the facial expression while ignoring the centrally overlaid emotional word. Congruent trials feature matching facial and semantic emotions (e.g., a happy face overlaid with the word

"HAPPY"). In contrast, incongruent trials feature conflicting cross-modal emotional information (e.g., an angry face overlaid with the word "HAPPY"), which is designed to elicit the emotional interference effect. Panel B illustrates the experimental flow. Each trial began with a 1000 ms fixation interval, followed by stimulus presentation with a response window of up to 3000 ms. Participants categorize the target emotion as either angry (by pressing the 'A' key) or happy (by pressing the 'L' key). The flow example also depicts the face-target block; in the word-target block, participants instead classified the word while ignoring the face. Face stimuli come from the Radboud Faces Database (Langner et al., 2010).

Self-Report Questionnaires

Participants also completed a battery of validated questionnaires. The present thesis used only three of the collected measures, values of which are reported in Table 1. The broader details regarding the questionnaires are reported in the study protocol by Morssinkhof et al. (2025a), with data-collection context in Supplementary Data Collection Details.

Depressive symptoms were assessed via the 20-item Center for Epidemiologic Studies Depression Scale (CES-D; Radloff, 1977). CES-D asks about past-week frequency of symptoms on a 4-point Likert scale from 0 (*rarely or none of the time*) to 3 (*most or all of the time*). Total scores range 0 to 60, with higher scores indicating greater depressive severity (Radloff, 1977). In the analyzed sample, internal consistency was excellent (α = .904; George & Mallery, 2003), and baseline-to-follow-up correlation very strong (r = .883; Akoglu, 2018).

For descriptive purposes, emotion reactivity was measured with the 6-item Brief Emotion Reactivity Scale (B-ERS; Veilleux et al., 2023). Items were rated from 0 (*not at all like me*) to 4 (*completely like me*), producing total scores ranging from 0 to 24, where higher scores reflect greater emotional reactivity and more difficulty recovering from affective arousal. Internal consistency was good (α = .819; George & Mallery, 2003), and baseline-to-three-months correlation was strong (r = .634; Akoglu, 2018).

Autistic traits were measured with the 10-item Autism Spectrum Quotient-10 (AQ10; Allison et al., 2012), to contextualize the sample's neurodivergent profile. Items were rated from 1 (*definitely agree*) to 4 (*definitely disagree*) and reverse-coded where appropriate, yielding total scores from 10 to 40; higher scores indicate greater endorsement of autistic traits. Internal consistency was acceptable (α = .783; George & Mallery, 2003), and baseline-to-three-months correlation was almost very strong (r = .799; Akoglu, 2018).

Procedure

Data were collected from June 2025 to May 2026 at two timepoints: the day of GAHT initiation and three months later. At each visit participants completed remote self-report questionnaires via Castor Electronic Data Capture (EDC; 2019) before—or on the day of—the in-person behavioral assessment. Behavioral testing occurred at Amsterdam UMC on a standardized offline laptop in a quiet room. For the Face-Word Emotional Stroop Task participants completed a brief practice (two congruent, two incongruent trials; excluded from analysis) and then the counterbalanced experimental face-target and word-target blocks. Additional procedural details are provided in the Supplementary Data Collection Details.

Data Processing and Analyses

Data from the FWEST paradigm and questionnaires were processed and analyzed using R (v4.5.2; R Core Team, 2024) in RStudio interface (v2024.03.1; Posit Team, 2024). The confirmatory analysis plan was preregistered and is available at open science framework (OSF): https://osf.io/3fpt9/. Any deviations from preregistration are reported below.

Data Cleaning

Trial-level FWEST data were filtered according to the preregistered exclusion criteria. Trials without a response or with RT <200 milliseconds (ms) were excluded, whereas inaccurate responses were retained as valid RT trials. The initially merged dataset contained the maximum possible 3,536 trials (136 trials × 2 timepoints × 13 participants) prior to applying exclusions. After exclusion, 3,493 trials (98.8%) remained; 43 trials were removed, including 42 non-responses and one response below 200 ms. All 13 participants met the preregistered data-quality thresholds (>75% valid trials and >50% accuracy) and were retained in the confirmatory and exploratory analyses. For the CES-D, three participants missed one baseline item, and two participants missed one item at the 3-month follow-up. Subsequently, missing CES-D item responses were coded as missing and handled as missing in subsequent analyses (i.e., no imputation employed). There were no missing values for the B-ERS or AQ10 items.

Deviation from Preregistered Analysis and the Adapted Analytic Strategy

The preregistered plan expected running a maximally specified, trial-level Bayesian mixed-effects bivariate model (i.e., jointly modelling RT and accuracy [ACC]). The a priori sensitivity analysis assumed an analytic sample of N = 30 (see Supplementary Materials Power Analysis Details), approximated to provide 80% power to detect minimum standardized effect sizes (SES) of 0.56 for the directional main effect of emotional interference (RQ1) and SES of 0.63 for the non-directional interaction effects (RQ2–RQ4); the corresponding sensitivity analysis is shown in Supplementary Figure S3.

However, initial inspection of the data showed near-ceiling ACC across several conditions, leaving insufficient variance for joint RT and ACC modelling; ACC was therefore excluded from further analysis. In addition, the achieved group sizes (n = 8 and n = 5) provided too little information to estimate complex (sub)group-level hierarchical variance components (McNeish & Stapleton, 2014). As a result, the maximally specified RT mixed-effects model—preserving the preregistered fixed- and random-effects structure and estimated separately within each GAHT group—failed to converge reliably.

Consequently, the analytical strategy was adapted to the Split/Analyze/Meta-Analyze (SAM) Bayesian framework (Cheung & Jak, 2016). In this SAM approach, each participant was first modelled separately, yielding participant-specific and corresponding uncertainty estimates, which were then meta-analytically synthesized within the M-GAHT and F-GAHT groups separately. Simplified group-level RT models were retained as sensitivity analyses using the same fixed-effect structure as the individual-level SAM models, enabling appropriate comparison.

Individual-Level Reaction-Time Mixed-Effects Model

In the Split and Analyze phases, individual-level Bayesian RT mixed-effects models were fitted for each participant across both timepoints using *brms* with the *rstan* backend (Bürkner, 2021; Stan Development Team, 2025). RTs were modelled using a shifted lognormal likelihood to accommodate their positive skew (Ulrich & Miller, 1993). A lognormal shifted family was selected over Gaussian and ex-Gaussian using data-driven leave-one-out cross-validation and posterior predictive checks (Vehtari et al., 2017). Models used four Monte Carlo chains with 4,000 iterations per chain (1,000 warmup) and adapt_delta = 0.95 (Margossian & Gelman, 2023; Stan Development Team, 2025).

All binary categorical predictors were coded using deviation contrasts (−0.5/+0.5), so each coefficient represents the full difference between the two factor levels (see Modelling Contrasts in the Supplementary Materials for details). The participant-level model syntax was: *rt_ms ~ congruency * Timepoint * distractor + block + stim_gender + (1 | SID).* Here, rt_ms denotes trial-level RT in ms; congruency compares congruent and incongruent trials; Timepoint compares baseline and three-month assessments; and distractor compares happy and angry distractors. The interaction term therefore tests whether the congruency effect, that is, emotional interference, differs by distractor valence and changes over time. The block term controlled for whether the target was a face or a word and stim_gender controlled for stimulus gender. Finally, (1 | SID) included a random intercept for stimulus identity, accounting for repeated observations of the same stimuli.

The confirmatory parameters were interpreted via SES and the preregistered region of practical equivalence (ROPE) criteria (Kruschke, 2018). The congruency term indexed general emotional interference (RQ1/H1), with positive values indicating slower RTs on incongruent than congruent trials. The congruency:distractor indexed valence-specific emotional interference bias (RQ2/H2), with positive values indicating stronger angry- than happy-distractor interference and negative values indicating stronger happy- than angry-distractor interference. The term congruency:Timepoint indexed longitudinal change in general emotional interference (RQ3/H3), with positive values indicating increased general emotional interference at three months and negative values indicating reduced general emotional interference. The three-way congruency:Timepoint:distractor term indexed longitudinal change in valence-specific emotional interference bias (RQ4/H4), with positive values indicating a shift toward stronger angry-relative-to-happy distractor interference and negative values indicating a shift toward stronger happy-relative-to-angry distractor interference. Estimates for H1–H2 were visualized in Figure 3 and H3–H4 estimates in Figure 4 using forest plots of participant-level posterior medians (i.e., SES), 95% highest-density intervals (HDIs), and group meta-analytic SES.

Bayesian Meta-Analysis

In the Meta-Analyze phase, participant-level SES estimates and their standard errors were synthesized via Bayesian random-effects meta-analysis: *SES_Mean | se(SES_SD) ~ 1 + (1 | Participant_ID).* The intercept estimated the group-level SES, and the participant-level random effect estimated between-person heterogeneity. Models were fitted separately for M-GAHT and F-GAHT using a Student-t(3, 0, 1) intercept prior, a Cauchy(0, 0.5) random-effect standard deviation prior, four chains, 3,000 iterations, 1,000 warmup iterations, and adapt_delta = 0.99 (Stan Development Team, 2025). By weighting each participant's effect by its posterior uncertainty, as opposed to treating all participants as contributing equally to group estimates, the SAM model produced precision-weighted group estimates. The SAM framework therefore avoided the convergence failures, and the potentially spuriously narrow uncertainty intervals characteristic of fully pooled mixed-effects models fitted to very small groups (McNeish & Stapleton, 2014; Vehtari et al., 2021).

Group-Level Sensitivity Analysis

After the SAM analyses, group-level RT mixed-effects models were fitted as sensitivity checks using the same fixed-effects structure and shifted lognormal family: *rt_ms ~ congruency * Timepoint * distractor + block + stim_gender + (1 | Participant_ID) + (1 | SID).* This model estimated pooled trial-level fixed effects while accounting for participant-level clustering and repeated presentation of the same facial identities. Differences between SAM and group-level estimates may therefore reflect differences in weighting, standardization, shrinkage, and between-person heterogeneity assumptions. Additional group-level sensitivity variants including age, previous-trial congruency, and maximally specified random effects did not converge satisfactorily (R-hat > 1.05; Vehtari et al., 2021), and were therefore not interpreted nor reported below.

Exploratory CES-D Analyses

The first exploratory group-level Bayesian RT mixed-effects model tested whether depressive symptom severity was associated with general emotional interference within each GAHT group. CES-D total scores were grand-mean z-standardized, and the distractor-valence model was specified as: *rt_ms ~ congruency * CESD_Z + Timepoint + block + distractor + stim_gender + (1 | SID) + (1 | Participant_ID).* The parameter of interest was the congruency:CESD_Z interaction. Under the contrast coding used (incongruent = +0.5, congruent = –0.5), positive SES values indicate stronger emotional interference at higher CES-D scores, whereas negative values indicate weaker interference at higher CES-D scores. This association was visualized descriptively by coloring each participant's H1 estimates by overall CES-D score (Figure 5).

The second exploratory group-level Bayesian RT mixed-effects model tested whether within-person CES-D change was associated with within-person change in general emotional interference within each GAHT group. CES-D change was computed as 3-month CES-D minus baseline CES-D and then z-standardized across participants, with positive values indicating symptom increase over time. The distractor-valence model was specified as: *rt_ms ~ congruency * Timepoint * CESD_Change_Z + block + distractor + stim_gender + (1 | SID) + (1 | Participant_ID)*, where main congruency:Timepoint:CESD_Change_Z, parameter positive SES values indicate that larger within-person CES-D increases are associated with greater increases in emotional interference over time, whereas negative values indicate that larger CES-D increases are associated with greater reductions in emotional interference. The participant-level emotional interference slopes from baseline to three months were plotted and color-coded by within-person CES-D change to allow visual evaluation of emerging patterns (Figure 6).

Model Convergence Diagnostics

Model convergence and posterior sampling quality were assessed using rank-normalized R-hat values, bulk and tail effective sample sizes, checks for divergent transitions, trace plots, and posterior predictive checks (Stan Development Team, 2025). Estimates were interpreted only if converged at R-hat ≤ 1.05, with those reported below falling ≤ 1.01, following recommendations for robust convergence diagnostics (Vehtari et al., 2021). Trace plots were inspected for chain mixing, and posterior predictive checks were used to evaluate whether the shifted-lognormal likelihood adequately reproduced the observed RT distribution. Results were reported only for models that met both numerical and visual diagnostic criteria.

Evidence Evaluation Criteria

Posterior fixed-effect draws were standardized by dividing each coefficient draw by the posterior residual standard deviation (σ), yielding standardized effect-size (SES) draws, as preregistered. Posterior distributions were summarized using the median SES and 95% HDIs. Evidence was evaluated against the preregistered ROPE = [−0.1, +0.1] SES, following the HDI + ROPE decision framework (Kruschke, 2018). This ROPE threshold ensured that trivially small effects just above or below zero were not treated as significant. Effects were classified as practically significant when the full 95% HDI lay outside the ROPE, practically equivalent when the full HDI fell within the ROPE, and inconclusive when the HDI overlapped either ROPE boundary. Estimates whose HDIs were entirely positive or entirely negative relative to zero but still crossed a ROPE boundary were described as directional and were otherwise classified as ROPE-inconclusive when the 95% HDI also crossed zero. All confirmatory and exploratory questions were examined per person and at the GAHT group level.

Results

It was first examined whether participants exhibit the presence of general emotional interference (RQ1). For visualization of individual and group-level meta-analytic estimates refer to Figure 3 (left panel). Only two participants showed practically significant positive effects, with 95% HDIs entirely above the ROPE: P06 in the M-GAHT group (SES = 0.68, 95% HDI [0.44, 0.94]) and P02 in the F-GAHT group (SES = 0.42, 95% HDI [0.13, 0.68]). Four additional participants had HDIs entirely above zero but overlapping the ROPE, whereas the remaining seven had positive medians with HDIs crossing zero. At the group level, the M-GAHT aggregate was practically significant in the positive direction (SES = 0.26, 95% HDI [0.11, 0.42]), whereas the F-GAHT aggregate estimate was directionally positive but not practically significant (SES = 0.21, 95% HDI [0.05, 0.38]).

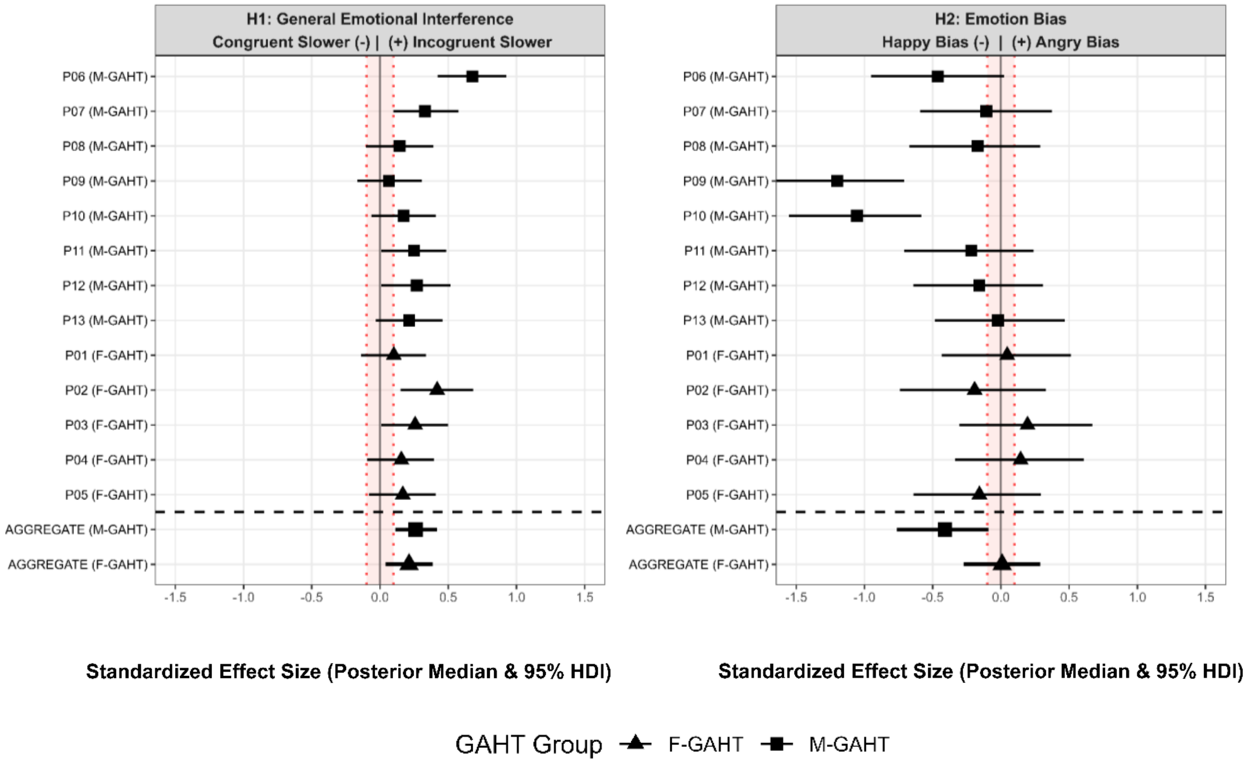
When assessing estimates for valence-specific emotional interference bias (see visualized in Figure 3, right panel), two M-GAHT participants showed practically significant negative effects: P09 (SES = −1.20, 95% HDI [−1.69, −0.71]) and P10 (SES = −1.06, 95% HDI [−1.55, −0.58]). No other participant showed a directional negative or positive person-level estimate. At the group level, the M-GAHT aggregate estimate was negative but only directional (SES = −0.41, 95% HDI [−0.76, −0.09]), whereas the F-GAHT aggregate remained ROPE-inconclusive centered at zero.

Longitudinal estimates for emotional interference (Figure 4, left panel) showed no practically significant baseline-to-three-month change at the individual or GAHT group level. Most participant-level medians clustered near zero, with intervals spanning both negative and positive values. Two M-GAHT participants showed directional negative estimates, suggesting possible reduced interference over time, with no directional evidence leaning in the opposite positive direction. Both group aggregate estimates were centered at and overlapped zero and were thus ROPE inconclusive.

Valence-specific longitudinal estimates (Figure 4, right panel) had the more widely distributed 95% HDIs. Two participants showed practically meaningful effects in opposite directions: P11 from the M-GAHT group showed a negative effect (SES = −1.64, 95% HDI [−2.63, −0.70]), whereas P05 from the F-GAHT group showed a positive effect (SES = 1.17, 95% HDI [0.19, 2.09]). The remaining 11 participants showed ROPE-inconclusive estimates, with medians scattered on both sides of zero. At the group level, both the M-GAHT and F-GAHT aggregates median overlapped zero. Full estimates are in Supplementary Table S1.

Figure 3

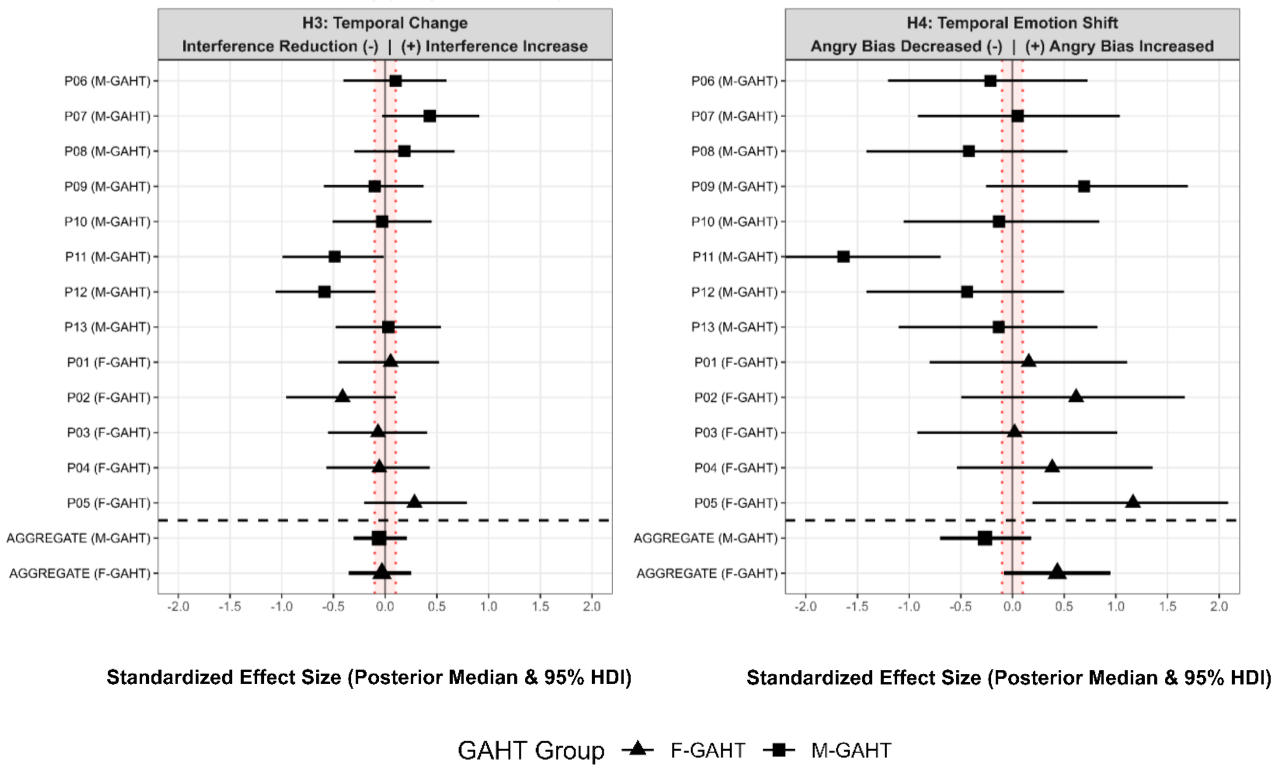
Posterior Estimates of General Emotional Interference and Distractor-Valence Bias



Note. F-GAHT = feminizing gender-affirming hormone therapy; M-GAHT = masculinizing gender-affirming hormone therapy; SES = standardized effect size; HDI = highest density interval; ROPE = Region of Practical Equivalence. Forest plots display individual posterior median SES values and 95% HDIs and meta-analytic group estimates from Bayesian mixed-effects models. The left panel H1 shows general emotional interference (positive values indicate slower reaction times on incongruent than congruent trials and negative values indicate faster responses on incongruent than congruent trials). The right panel H2 shows emotional interference by distractor valence (negative values indicate stronger interference by happy than angry distractors and positive values indicate stronger interference by angry than happy distractors). The red dotted vertical lines and shaded band denoting the preregistered region of practical equivalence (ROPE) was [−0.10, +0.10] SES. Squares denote M-GAHT while triangles F-GAHT participants. Group estimates are plotted below the dashed line.

Figure 4

Posterior Estimates of Longitudinal Change in General Emotional Interference and Longitudinal Change in Distractor-Valence Bias



Note. F-GAHT = feminizing gender-affirming hormone therapy; M-GAHT = masculinizing gender-affirming hormone therapy; SES = standardized effect size; HDI = highest density interval; ROPE = region of practical equivalence. Forest plots display individual posterior median SES values and 95% HDIs and meta-analytic group estimates from Bayesian mixed-effects models. The left panel H3 shows change in general emotional interference from baseline to three months (positive values indicate increased and negative reduced interference). The right panel H4 shows longitudinal change in interference by distractor valence (positive values indicate a shift toward stronger angry-relative-to-happy distractor interference and negative values indicate a shift toward stronger happy-relative-to-angry distractor interference). The solid black vertical line marks zero. The red dotted vertical lines and shaded band denoting the preregistered region of practical equivalence (ROPE) was [−0.1, +0.1] SES. Squares denote M-GAHT while triangles F-GAHT participants. Group estimates are plotted below the dashed line.

Group-Level Sensitivity Analysis

The group-level RT models were used solely as practical-significance sensitivity checks for the SAM-derived group estimates. For general emotional interference (H1), the sensitivity models yielded practically significant positive effects in both groups: M-GAHT (SES = 0.24, 95% HDI [0.16, 0.33]) and F-GAHT (SES = 0.23, 95% HDI [0.12, 0.33]). For valence-specific emotional interference bias (H2), the M-GAHT group showed a practically significant negative estimate (SES = −0.39, 95% HDI [−0.56, −0.21]), whereas the F-GAHT group estimate overlapped zero. For longitudinal change in general emotional interference (H3), both groups overlapped zero. For longitudinal change in valence-specific emotional interference bias (H4), the M-GAHT estimate overlapped zero, whereas the F-GAHT estimate

(SES = 0.51, 95% HDI [0.10, 0.95]) was positive but borderline with respect to the ROPE boundary. Full results are reported in Supplementary Table S2.

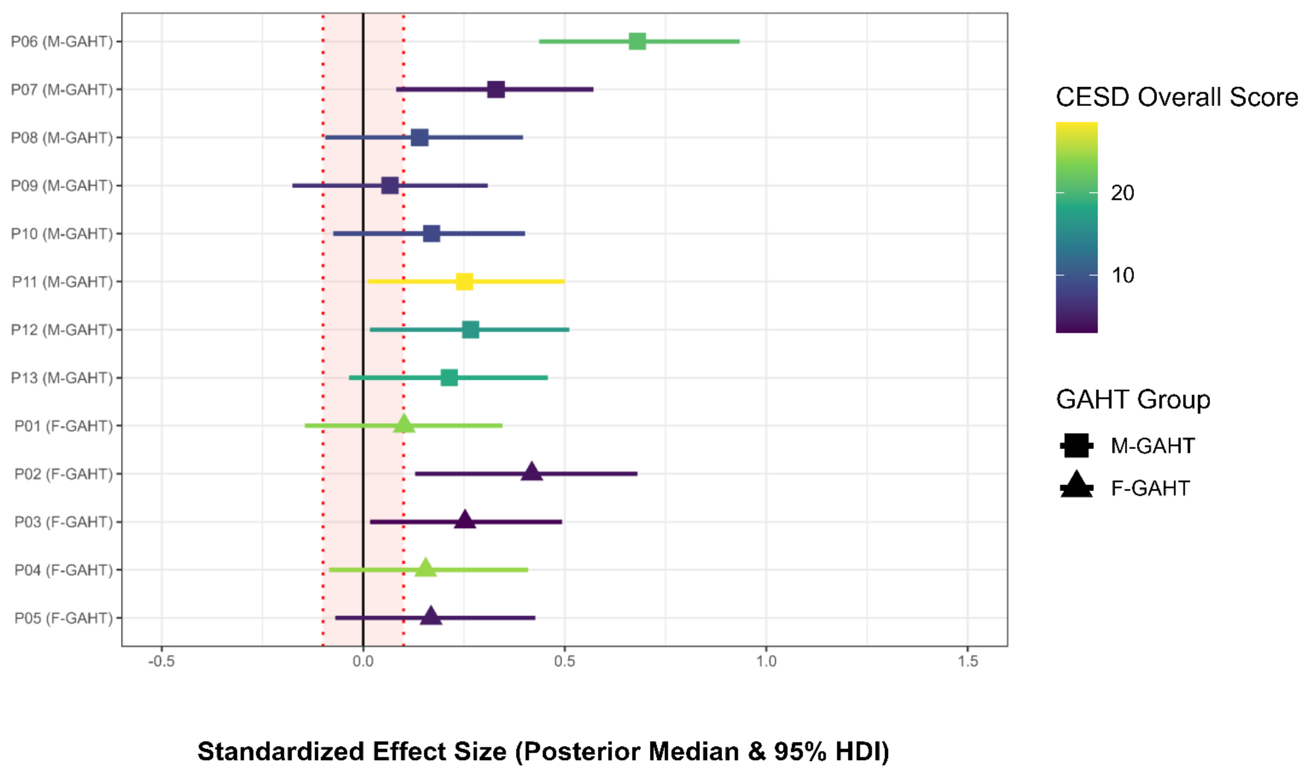
Depressive Symptomatology and Emotional Interference Associations

To assess whether depressive symptom severity was associated with general emotional interference (RQ5a), individual general emotional interference estimates were first visualized by overall CES-D score per individual (Figure 5). The color CES-D gradient did not track the ordering of interference estimates consistently. One participant with comparatively elevated depressive symptoms had the largest interference estimate, but the participant with the highest CES-D score fell within the range of several lower-symptom participants. At the group model level (see Supplementary Table S3), neither M-GAHT nor F-GAHT showed a practically meaningful association of CES-D with general emotional interference. The M-GAHT estimate was directionally positive with lower bound exactly at zero; the F-GAHT group estimate crossing zero completely.

Lastly, it was examined whether within-person change in depressive symptoms was associated with within-person change in general emotional interference (RQ5b; Figure 6). In the M-GAHT group, four participants showed increased general emotional interference; three of which also showed increases in CES-D scores over time, while one showed only a little CES-D decrease. Among the four M-GAHT participants whose emotional interference decreased over time, two showed slight increase, and two slight decreases in CES-D scores. In the F-GAHT group, trajectories were more heterogeneous, with three participants showing decreased general emotional interference, accompanied by divergent CES-D changes, with two showing increases in depressive symptoms and one showing a decrease. Two participants showed increased general emotional interference, with one displaying a large CES-D increase despite only a small interference increase and the other showing a small CES-D increase despite a moderate interference increase. At the group model level, both M-GAHT and F-GAHT change-change estimates overlapped zero (see Supplementary Table S3).

Figure 5

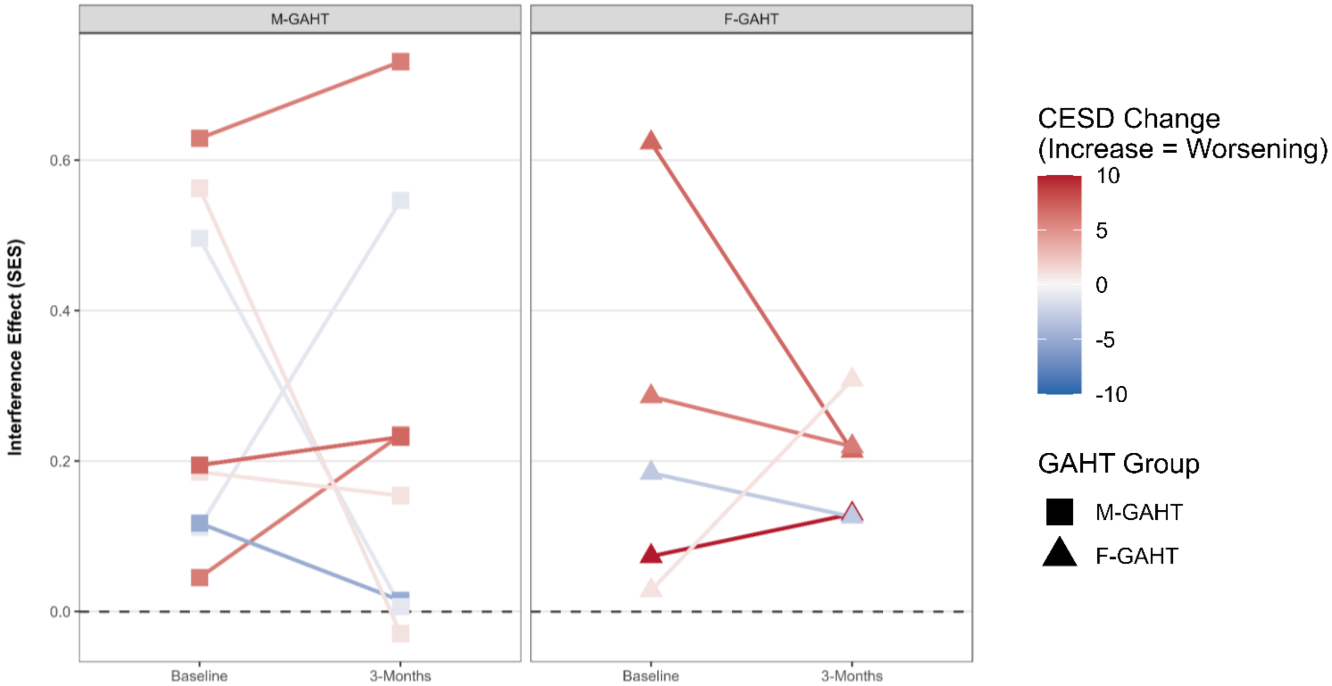
General Emotional Interference Estimates Colored by Overall Depressive Symptom Severity



Note. F-GAHT = feminizing gender-affirming hormone therapy; M-GAHT = masculinizing gender-affirming hormone therapy; CES-D = Center for Epidemiologic Studies Depression Scale; SES = standardized effect size; HDI = highest density interval; ROPE = region of practical equivalence. The forest plot displays participant-level posterior median SES values and 95% HDIs for general emotional interference. Point and interval color represents each participant's overall CES-D score, with higher values indicating higher depressive symptom severity. Positive SES values indicate slower reaction time on incongruent than congruent trials. The red dotted vertical lines and shaded band denote the preregistered ROPE [−0.1, +0.1] SES. Squares denote M-GAHT while triangles F-GAHT participants.

Figure 6

Within-Subject Trajectories of General Emotional Interference From Baseline to Three Months Colored by Within-Subject CES-D Change



Note. F-GAHT = feminizing gender-affirming hormone therapy; M-GAHT = masculinizing gender-affirming hormone therapy; CES-D = Center for Epidemiologic Studies Depression Scale; SES = standardized effect size. Participant-level slopes display general emotional interference estimates from baseline to three months. Line color represents within-person CES-D change over the same interval, computed as three-month CES-D minus baseline CES-D. Positive CES-D change values indicate increased depressive symptoms and are shown in red hues; negative values indicate decreased depressive symptoms and are shown in blue hues; values near zero are shown near the neutral midpoint. Squares denote M-GAHT and triangles F-GAHT participants. The dashed horizontal line marks zero emotional interference.

Discussion

The present thesis examined emotional interference in 13 TGD adults initiating GAHT, including five participants initiating F-GAHT and eight initiating M-GAHT, with analyses conducted at the individual level and summarized within each GAHT group. The primary questions were whether participants showed general emotional interference, defined as slower responding to emotionally incongruent than congruent face-word information (RQ1); whether this interference was biased toward angry versus happy distractors (RQ2); whether general emotional interference changed from baseline to three months after GAHT initiation (RQ3); and whether valence-specific emotional interference bias shifted over the same period (RQ4). Exploratory analyses further examined whether overall depressive symptom severity was associated with general emotional interference (RQ5a), and whether within-person depressive-symptom change co-occurred with within-person change in general emotional interference (RQ5b).

H1 was partially supported, with the clearest evidence in the M-GAHT group and more tentative, directionally consistent evidence in the F-GAHT group. At the individual level, however, general emotional interference was not robustly expressed, as most participant estimates remained at trend level. Twelve out of 13 estimates were smaller than the pooled effect in Supplementary Figure S1, suggesting that general emotional interference in this longitudinal GAHT context may be weaker or less consistently expressed than in prior experimental samples (Fan et al., 2016; Song et al., 2017). However, the task appeared to elicit the expected effect size of general emotional interference effect with group SES = 0.26 for M-GAHT and SES = 0.21 for F-GAHT, closely matching the effect reported in the FWEST study on which the present task was based (d = 0.26; Agustí et al., 2017).

Evidence for valence-specific emotional interference bias was more heterogeneous. Despite the theoretical possibility that minority stress, chronic expectations of hostility, and vigilance to social threat might be reflected in stronger threat-related distractibility (Fox et al., 2000; Kiyar et al., 2022b; Meyer, 2003; Puckett et al., 2023), no participant showed a practically significant angry-distractor bias. Instead, two M-GAHT participants, P09 and P10, showed clear happy-distractor-biased interference. The SAM estimate therefore showed a directional negative trend, and the group-level sensitivity model showed a practically significant negative effect in the M-GAHT group. However, because this pattern was driven by two large person-level SES estimates (SES = −1.20 and SES = −1.06, respectively) and was not evident in the remaining five M-GAHT participants, it is better interpreted as an example of how individual estimates can pull group averages toward apparent significance, rather than as evidence that M-GAHT generally induces preferential distraction by happy stimuli (see Figure 3, right panel, and Supplementary Table S2).

Evidence for longitudinal change in general emotional interference was inconclusive. Although it is highly plausible that the first months of GAHT are life phase in which affective experience, psychosocial functioning, and social-emotional processing may begin to change (Doyle et al., 2023; Kiyar et al., 2022a; Matthys et al., 2021; Morssinkhof et al., 2025b), neither the SAM estimates nor the group-level sensitivity models indicated a systematic increase or decrease in general emotional interference from baseline to three months. Individual trajectories were largely heterogeneous; two M-GAHT participants showed directionally negative within-person estimates, suggesting a possible reduction in interference over time, but there was no corresponding directional pattern of evidence at the M-GAHT group-level. Thus, the present data do not suggest that early GAHT was accompanied by a large shift in the general RT cost of resolving incongruent emotional face-word information.

Evidence for longitudinal change in valence-specific emotional interference bias was also inconclusive at the SAM group level. This last confirmatory question tested whether the relative interference from angry versus happy distractors shifted from baseline to three months. Two participants showed opposing patterns: P11 in the M-GAHT group shifted toward stronger happy-distractor interference, whereas P05 in the F-GAHT group shifted toward stronger angry-distractor interference. Although the F-GAHT sensitivity model yielded a positive estimate at the ROPE boundary, the SAM-level uncertainty and largely imprecise participant-level estimates suggest that this was not robust evidence for F-GAHT group-wide valence-specific change toward angry-biased emotional interference. As with the H2 valence-bias results, the discrepancy between SAM and group-level sensitivity results likely reflects a pooled signal in a small, heterogeneous sample, where group-level estimates can be strongly shaped by few large individual effects (McNeish & Stapleton, 2014).

The exploratory depressive-symptom analyses did not provide reliable evidence for a consistent association between depressive symptomatology and general emotional interference. Numerically, no practically significant association was observed between overall depressive severity and general emotional interference in either GAHT group, and visual inspection did not show a systematic pattern (see Figure 5 and Supplementary Table S3). This finding did not align with meta-analytic evidence that depressive symptomatology is associated with stronger emotional Stroop interference (Epp et al., 2012), but should be interpreted cautiously given the small TGD sample initiating GAHT.

We then asked whether within-person changes in depressive symptoms co-occurred with within-person changes in general emotional interference from baseline to three months. Visual inspection suggested a tentative M-GAHT pattern, where several participants with increased depressive symptoms also showed increased emotional interference over time (see Figure 6, left panel). However, this pattern was not mirrored for symptom decreases, did not appear systematically in the F-GAHT group, and was not supported by the group model estimates (Supplementary Table S3). Accordingly, any apparent link between rising depressive symptoms and greater emotional interference should be treated as provisional.

Population Description

Descriptively, as shown in Table 1, CES-D scores increased from baseline to three months in both GAHT groups, numerically more in F-GAHT (+4.20) than in M-GAHT (+1.75), although group differences were not tested inferentially. This overall increase contrasts with systematic evidence that depressive symptoms and psychological distress often improve after GAHT (Doyle et al., 2023; Van Leerdam et al., 2023; White Hughto & Reisner, 2016). The larger descriptive increase in F-GAHT also parallels Morssinkhof et al. (2024), who reported regimen-differentiated depressive-symptom trajectories, with modest low-mood worsening after feminizing but not masculinizing GAHT. In contrast, B-ERS scores showed the opposite relative pattern to CES-D, increasing in M-GAHT (+1.00) but decreasing slightly in F-GAHT (−0.40), whereas AQ10 scores remained largely stable, consistent with autistic traits being relatively trait-like (Lord et al., 2020).

Statistical and Methodological Implications

The broader message of these findings is that average GAHT-group estimates can be informative, but they are not the same as individual behavioral trajectories. In this study, the clearest apparent group-level signals often emerged because one or two participants showed relatively large effects, while the same pattern was absent in most others within the respective GAHT group. This was most visible for valence-specific bias in the M-GAHT group, where two participants pulled the group estimate toward apparent happy-distractor bias despite five of eight participants not showing that pattern. Without the individualized SAM approach, this could have been misread as evidence that M-GAHT generally induces happy-biased emotional interference, rather than as a participant-level pattern that became visible only because individual estimates were examined directly.

Therefore, this showcased the SAM approach was a conceptually appropriate adaptation of the preregistered group-level analysis plan, given the small sample size and the heterogeneity observed across TGD participants starting GAHT (Cheung & Jak, 2016; McNeish & Stapleton, 2014). More broadly, these findings remain agnostic about whether M-GAHT and F-GAHT are the most informative categories for explaining behavioral emotional-interference change. Although this distinction is clinically and theoretically useful, each group may still contain substantial variation in endocrine regimen, dose, route of administration, prior hormone history, embodiment goals, psychiatric history, neurodivergence, social context, and gender-affirming aims (Coleman et al., 2022; Hembree et al., 2017; Nguyen et al., 2018a, 2018b). For this reason, individualized approaches may be especially informative when studying early behavioral changes during GAHT. Collectively, the present findings suggest that change effects in general or valence-specific emotional interference, if present, may be subtle, heterogeneous, or difficult to detect with the current FWEST design during the first three months of treatment.

Clinical Implications

Moreover, the clinical interpretability of observed FWEST effects remains limited. The task indexes rapid emotional-conflict processing under constrained laboratory conditions (Etkin et al., 2006; Song et al., 2017), and may therefore be theoretically informative for studying cognitive-affective control (Inzlicht et al., 2015). However, most person-level estimates were small to moderate in standardized units (Cohen, 1988), corresponding roughly to a tenth of a second, depending on each participant's RT distribution and modelled contrast. As with many controlled laboratory paradigms, this supports experimental precision but limits ecological generalization to complex real-world social interaction (Shamay-Tsoory & Mendelsohn, 2019). For clinical purposes, it seems therefore much more informative to qualitatively assess how TGD individuals themselves describe changes in emotionality, communication, conflict management, avoidance, and social support (e.g., as in Millet et al., 2026), instead of attempting to infer interpersonal functioning from behavioral data alone.

Limitations and Future Directions

Several limitations apply to this study. First, the present models did not separate face-target and word-target blocks, even though prior face-word Stroop studies suggest that interference can depend on whether faces or words dominate task performance (Beall & Herbert, 2008; Fan et al., 2016; Ovaysikia et al., 2011). As a result, the present analyses cannot determine whether the observed happy-distractor patterns were driven mainly by happy faces, happy words, or their combination. Second, the task used a fixed happy-angry contrast, which was suitable for testing the present TGD population, but sadness may be more directly relevant for depression-focused emotional interference based on meta-analytic and empirical evidence (Başgöze et al., 2015; Dalili et al., 2014; Ros et al., 2021). Third, precision was limited by the small sample, the modest number of trials per visit, and the number of interaction terms, which constrained both group-level and individual-level estimation. Fourth, ACC in this study was near ceiling levels (i.e., 100%) in the present study, limiting its usefulness as an additional outcome and suggesting that future FWEST versions may need shorter stimulus-presentation to increase error variance. Finally, the three-month observation windows might have been short for capturing behavioral changes induced by early GAHT.

Future work should build on the present findings and limitations by modelling face-target and word-target blocks separately and by using a broader set of emotions that can dissociate threat-related from sadness-related emotional interference. Doing so would clarify whether the happy-distractor bias seen in two M-GAHT participants reflects a reproducible superiority of happy stimuli, given evidence that happy faces can receive preferential processing in related visual-search paradigms (Stuit et al., 2025). Methodologically, future studies should increase trial counts per condition, shorten the response window, for example to 1000 ms (Fan et al., 2016), to make it possible to model ACC as an outcome, and recruit larger samples while appropriately considering inter-individual heterogeneity. Longer and more frequent follow-up assessments would also allow future studies to test whether affective changes precede behavioral changes, whether behavioral changes emerge only after affective adjustment has occurred, or whether the two trajectories change independently during GAHT. Finally, behavioral trajectories should be related to testosterone and estradiol levels or dosing changes, social support, and minority stress, so that emotional interference findings can be interpreted within the broader endocrine and psychosocial context of GAHT (Coleman et al., 2022; Hembree et al., 2017). The longitudinal AFFIRM Relationships cohort will be well-positioned to test whether behavioral outcomes track broader psychosocial-functioning changes after GAHT initiation (Morssinkhof et al., 2025a).

Conclusion

This thesis provides the first prospective behavioral evidence of emotional interference in transgender and gender-diverse adults initiating gender-affirming hormone therapy. Only a few participants showed practically or directionally significant emotional interference or valence-specific distractor-bias patterns, and there was no clear evidence that these would change over the first three months of treatment. The main conclusion is that in small and/or heterogeneous samples, a handful of individual effects can pull aggregate estimates away from patterns observed in most participants, thus misrepresenting the (hormonal) group. The clinical relevance of the emotional Stroop paradigm remains questionable. Future research should triangulate behavioral metrics with larger samples, longer follow-ups, hormone-level data, and first-person accounts to determine when, for whom, and how social-emotional behavioral patterns relate to emotional change.

Author Contributions

Petr Šupa: Conceptualization of the thesis subproject, literature review, task implementation, data collection, data processing, statistical analysis, visualization, interpretation, and writing. *Margot W. L. Morssinkhof*: Conceptualization, project supervision, parent-study coordination, methodology, interpretation, review and editing. *Anna Tyborowska*: Conceptualization, supervision, methodology, interpretation, review and editing. *Bernd Figner*: Methodological and statistical supervision. *David M. Doyle*: Funding and conceptualization of the parent AFFIRM Relationships project, supervision, review and editing.

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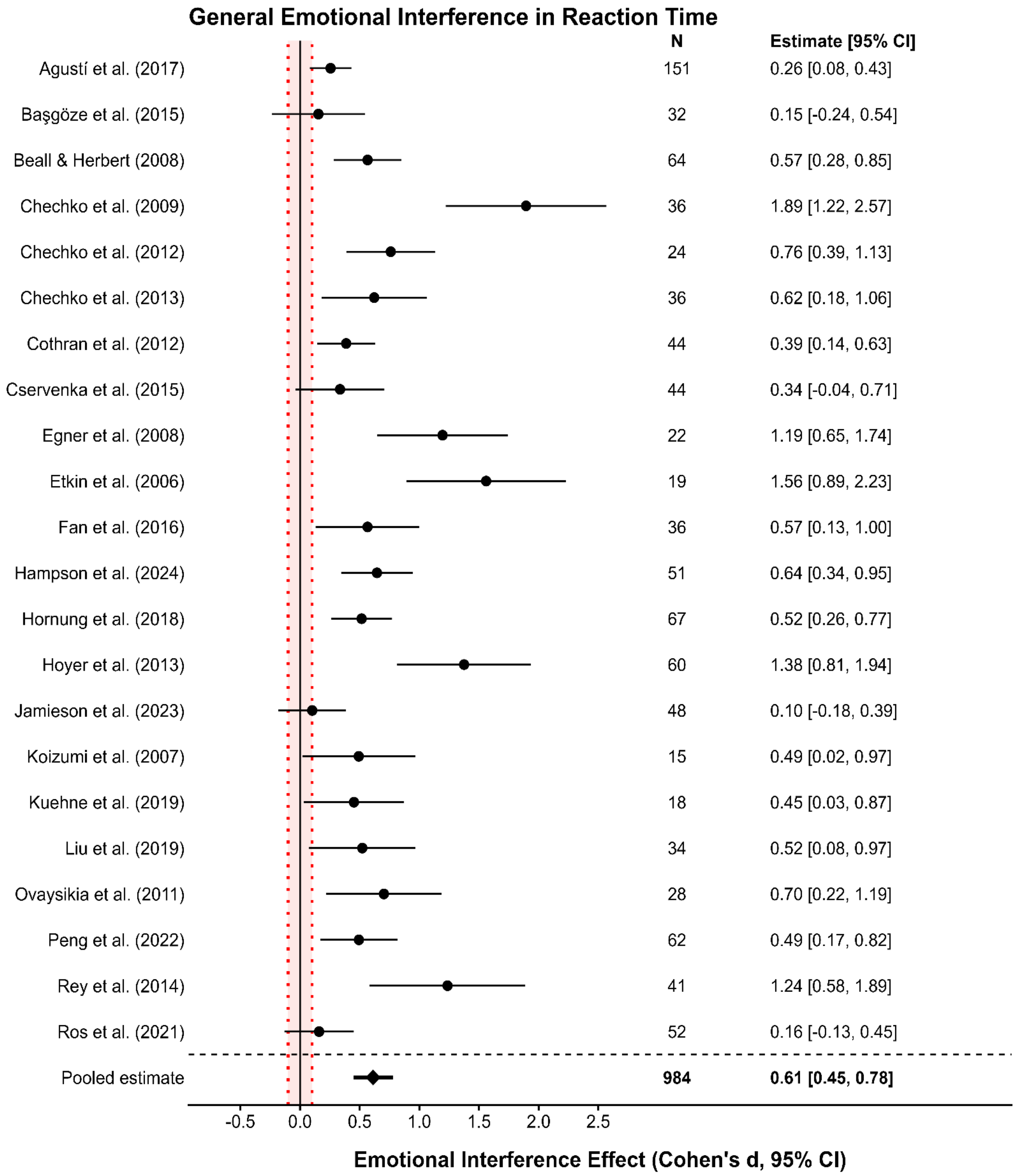
Supplementary Materials

Figure S1

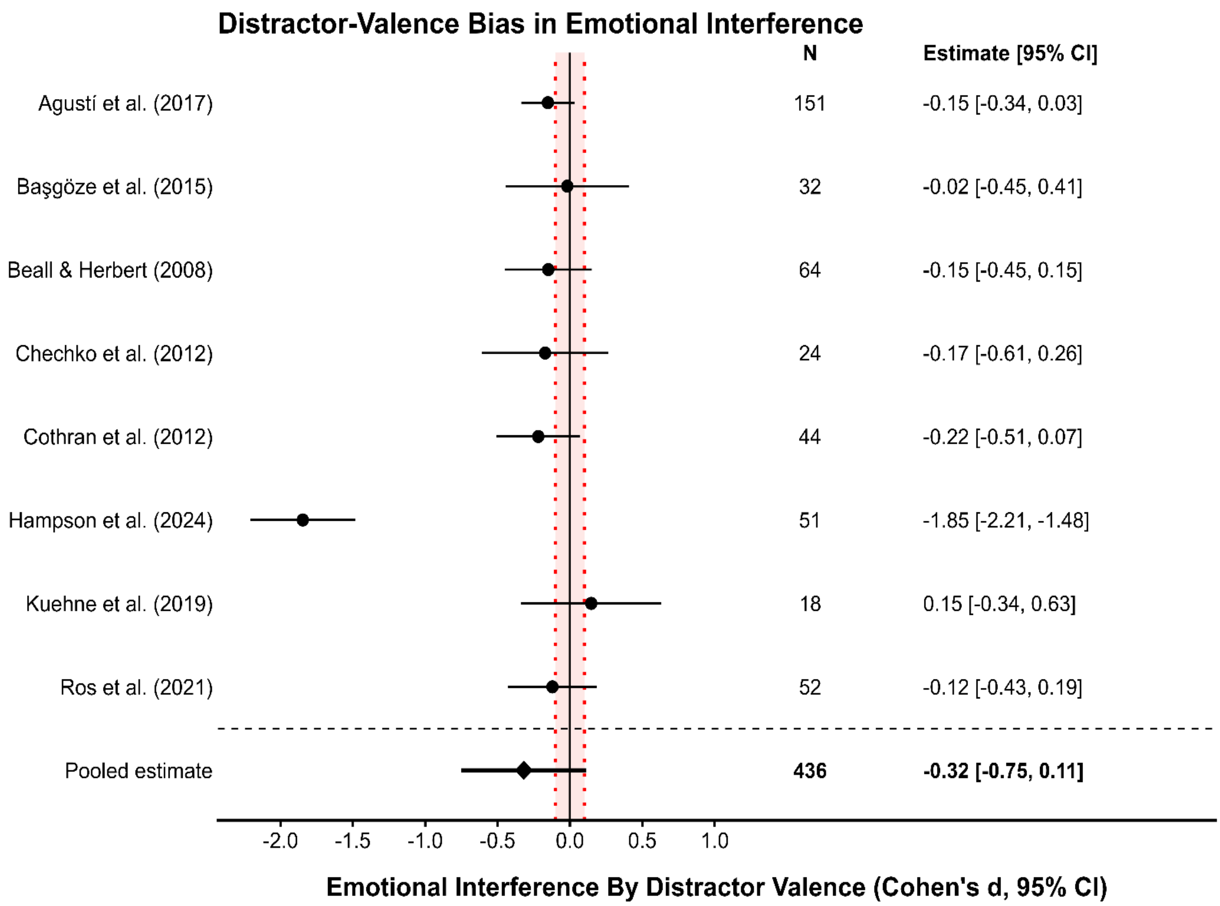
Meta-Analytic General Emotional Interference Evidence from Various Face-Word Emotional Stroop Task Paradigms



Note. Positive values indicate slower responding on incongruent than congruent trials. Points show study-level Cohen's *d* estimates with 95% confidence intervals (CI; Cohen, 1988); the diamond shows the pooled estimate from the meta-analytic random-effects model. The salmon-colored region of practical equivalence (ROPE) region marks [-0.1, 0.1] standardized effect-size units. This mini meta-analysis was used to contextualize the preregistered directional hypothesis 1 (H1) expectation and should not be understood as an exhaustive systematic review of all emotional Stroop variants.

Figure S2

Meta-Analytic Valence-Specific Emotional Interference Evidence from Various Face-Word Emotional Stroop Task Paradigms

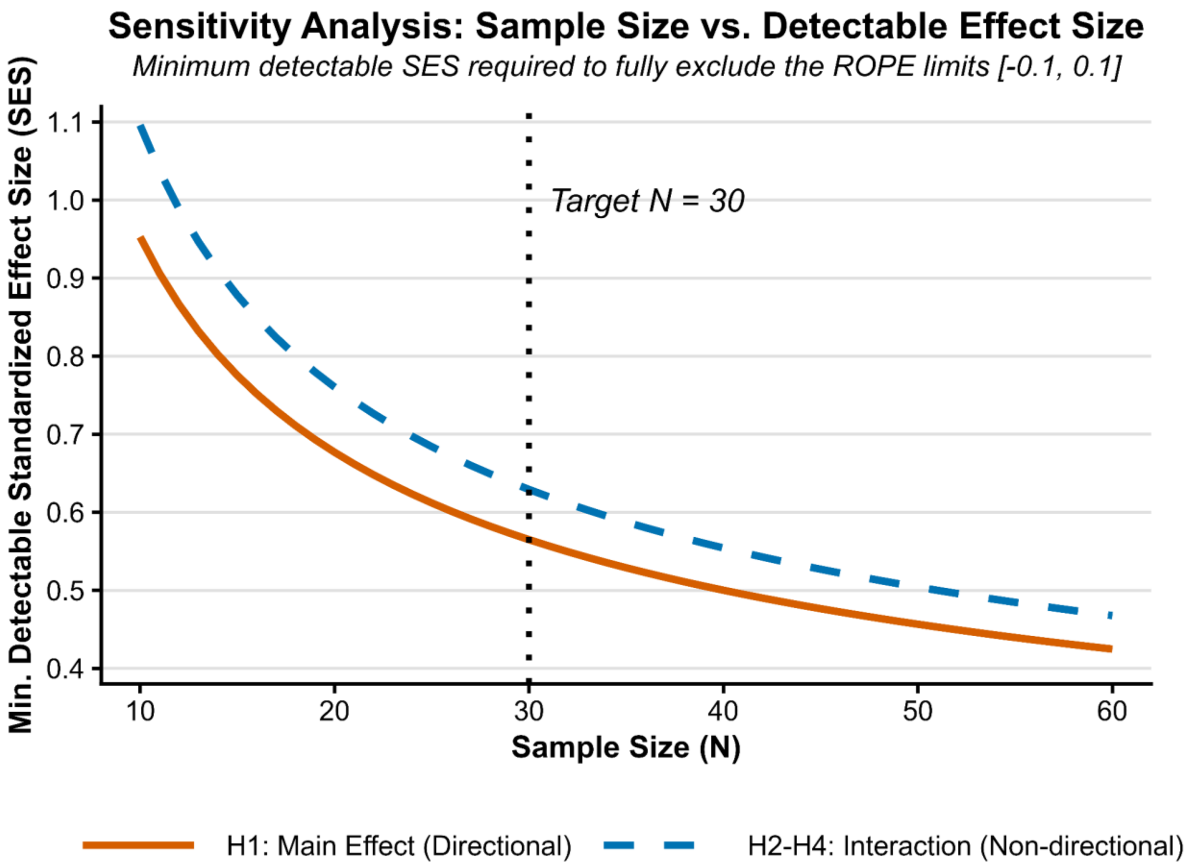


Note. Negative values indicate stronger interference for happy/positive distractors relative to angry/negative distractors; positive values indicate stronger angry/negative distractor interference. Points show study-level Cohen's *d* estimates with 95% confidence intervals (CI;

Cohen, 1988); the diamond shows the pooled estimate from the meta-analytic random-effects model. The salmon-colored region of practical equivalence (ROPE) region marks [-0.1, 0.1] standardized effect-size units. The wide confidence interval and heterogeneous study set motivated the non-directional specification of hypothesis 2 (H2).

Figure S3

Sensitivity Analysis of Sample Size and Minimum Detectable Standardized Effect Size



Note. Curves show the minimum standardized effect size (SES; Cohen's d_z paired-contrast approximation) required to achieve 80% power at α = .05 while exceeding the preregistered ROPE of [−0.1, 0.1] SES. The solid orange line represents the directional paired-contrast approximation for hypothesis 1 (H1), whereas the dashed blue line represents the non-directional paired-contrast approximation used to summarize H2–H4. The vertical dotted line marks the initially anticipated target sample size of $N = 30$, at which the minimum detectable SES was approximately $d_z = 0.56$ for H1 and $d_z = 0.63$ for H2–H4. The H2–H4 curve should be interpreted as a simplified sensitivity benchmark for Bayesian mixed-effects model power simulation; higher-order interaction terms, especially three-way interaction, may have lower practical sensitivity in the final trial-level Bayesian model.

Table S1

Individual and Aggregate Posterior Standardized Effect Size Estimates for Emotional Interference Effects and Longitudinal Changes Derived from Subject-Specific and Meta-Analytic Bayesian Mixed-Effects Models

Participant ID (GAHT)	H1: Emotional Interference	H2: Interference by Valence	H3: Interference by Time	H4: Interference by Time and Valence
P06 (M-GAHT)	**0.68 [0.44, 0.94]**	−0.46 [−0.95, 0.03]	0.10 [−0.40, 0.58]	−0.22 [−1.18, 0.77]
P07 (M-GAHT)	0.33 [0.08, 0.57]	−0.11 [−0.59, 0.37]	0.43 [−0.05, 0.93]	0.05 [−0.92, 1.04]
P08 (M-GAHT)	0.14 [−0.10, 0.40]	−0.17 [−0.67, 0.29]	0.19 [−0.32, 0.66]	−0.42 [−1.41, 0.53]
P09 (M-GAHT)	0.07 [−0.18, 0.31]	**−1.20 [−1.69, −0.71]**	−0.10 [−0.59, 0.37]	0.69 [−0.25, 1.66]
P10 (M-GAHT)	0.17 [−0.08, 0.40]	**−1.06 [−1.55, −0.58]**	−0.03 [−0.50, 0.45]	−0.13 [−1.05, 0.84]
P11 (M-GAHT)	0.25 [0.01, 0.50]	−0.22 [−0.71, 0.24]	−0.49 [−0.99, −0.02]	**−1.64 [−2.63, −0.70]**
P12 (M-GAHT)	0.27 [0.02, 0.51]	−0.16 [−0.64, 0.31]	−0.59 [−1.08, −0.08]	−0.44 [−1.41, 0.50]
P13 (M-GAHT)	0.21 [−0.04, 0.46]	−0.02 [−0.49, 0.47]	0.04 [−0.47, 0.55]	−0.13 [−1.10, 0.82]
P01 (F-GAHT)	0.10 [−0.15, 0.35]	0.05 [−0.43, 0.51]	0.06 [−0.44, 0.51]	0.16 [−0.80, 1.11]
P02 (F-GAHT)	**0.42 [0.13, 0.68]**	−0.19 [−0.74, 0.33]	−0.41 [−0.93, 0.14]	0.62 [−0.50, 1.67]

P03 (F-GAHT)	0.25 [0.02, 0.49]	0.20 [−0.31, 0.67]	−0.07 [−0.55, 0.40]	0.02 [−0.92, 1.01]
P04 (F-GAHT)	0.16 [−0.08, 0.41]	0.15 [−0.34, 0.61]	−0.06 [−0.56, 0.41]	0.39 [−0.54, 1.36]
P05 (F-GAHT)	0.17 [−0.07, 0.43]	−0.16 [−0.64, 0.29]	0.28 [−0.23, 0.78]	**1.17 [0.19, 2.09]**
M-GAHT (n = 8)	**0.26 [0.11, 0.42]**	−0.41 [−0.76, −0.09]	−0.06 [−0.30, 0.21]	−0.27 [−0.70, 0.18]
F-GAHT (n = 5)	0.21 [0.05, 0.38]	0.01 [−0.27, 0.29]	−0.03 [−0.34, 0.26]	0.43 [−0.08, 0.95]

Note. F-GAHT/M-GAHT = feminizing/masculinizing gender-affirming hormone therapy. Values are posterior median standardized effect sizes (SES) with 95% highest density intervals (HDIs). Practically significant estimates are in bold. Participant-level estimates were derived from subject-specific Bayesian shifted-lognormal mixed-effects models; GAHT group estimates are from Bayesian random-effects meta-analysis of participant-level SES estimates. SES were computed as fixed-effect coefficients divided by the posterior residual standard deviation (β/σ). Predictors were effect-coded, such that positive values indicate: H1, slower reaction times on incongruent than congruent trials; H2, stronger interference by angry than happy distractors; H3, increased interference from baseline-to-three-months; and H4, a longitudinal shift toward stronger angry-relative-to-happy distractor interference. Negative values indicate the reverse pattern. The preregistered region of practical equivalence (ROPE) was [−0.1, +0.1] SES. Effects were classified as practically significant only when the full 95% HDI fell outside the ROPE, practically equivalent when the full HDI fell within the ROPE, and inconclusive when the HDI overlapped the ROPE.

Table S2

Group-Level Bayesian Posterior Standardized Effect Size Estimates for Emotional Interference, Distractor-Valence Bias, and Their Longitudinal Changes

Group	H1: Emotional Interference	H2: Interference by Valence	H3: Interference by Time	H4: Interference by Time and Valence
M-GAHT (*n* = 8)	**0.24 [0.16, 0.33]**	**−0.39 [−0.56, −0.21]**	−0.10 [−0.27, 0.07]	−0.26 [−0.59, 0.08]
F-GAHT (*n* = 5)	**0.23 [0.12, 0.33]**	−0.06 [−0.29, 0.15]	−0.01 [−0.23, 0.21]	**0.51 [0.10, 0.95]**

Note. F-GAHT/M-GAHT = feminizing/masculinizing gender-affirming hormone therapy. Values are posterior median standardized effect sizes (SES) with 95% highest density intervals (HDIs). Practically significant estimates are in bold. Estimates are derived from the group-level Bayesian shifted-lognormal mixed-effects model. The model used the same fixed-effect distractor-valence structure and stimulus intercept as the subject-specific Split/Analyze/Meta-Analyze models reported in Table S1, with addition of random intercepts for participants. SES values were computed as fixed-effect coefficients divided by the posterior residual standard deviation. Predictors were effect-coded; positive estimates indicate: H1, stronger general emotional interference, defined as slower responses on incongruent than congruent trials; H2, stronger interference by angry than happy distractors; H3, increased emotional interference from baseline-to-three-months; and H4, a longitudinal shift toward stronger angry-relative-to-happy distractor interference. Negative estimates indicate the reverse pattern. The preregistered region of practical equivalence (ROPE) was [−0.1, +0.1] SES. Effects were interpreted as practically significant only when the full 95% HDI fell outside the ROPE, practically equivalent when the full HDI fell within the ROPE, and inconclusive when the HDI overlapped the ROPE.

Table S3

Group-Level Posterior Standardized Effect Size Estimates for CES-D Moderation of Emotional Interference and CES-D Change Moderation of Interference Change

Group	RQ5a: CES-D Severity Moderation of Emotional Interference	RQ5b: CES-D Change Moderation of Interference Change
M-GAHT (*n* = 8)	0.09 [0.00, 0.19]	−0.17 [−0.36, 0.03]
F-GAHT (*n* = 5)	−0.06 [−0.16, 0.04]	0.02 [−0.17, 0.21]

Note. F-GAHT/M-GAHT = feminizing/masculinizing gender-affirming hormone therapy; CES-D = Center for Epidemiologic Studies Depression Scale; SES = standardized effect size; HDI = highest density interval. Values are posterior median SES estimates with 95% HDIs from distractor-valence Bayesian shifted-lognormal mixed-effects models fitted separately within each GAHT group. For RQ5a, positive values indicate stronger general emotional interference at higher depressive symptom levels, whereas negative values indicate weaker interference at higher depressive symptom levels. For research question 5b (RQ5b), CES-D change was computed as 3-month CES-D minus baseline CES-D, and positive values indicate that larger increases in depressive symptoms were associated with greater increases in emotional interference over time, whereas negative values indicate that larger increases in depressive symptoms were associated with greater reductions in emotional interference over time. The preregistered region of practical equivalence (ROPE) was [−0.1, +0.1] SES. Effects were classified as practically significant only when the full 95% HDI fell outside the ROPE, practically equivalent when the full HDI fell within the ROPE, and inconclusive when the HDI overlapped the ROPE.

Data Collection Details

Participant data were obtained from two sources: (1) self-report questionnaires administered remotely through Castor EDC and (2) behavioral measures collected during in-person visits at the Center of Expertise on Gender Dysphoria at Amsterdam UMC. In-person visits were scheduled to coincide as closely as possible with participants' GAHT-related clinical appointments and blood draws, both to reduce participant burden and to enable future analyses linking hormonal measures to behavioral performance. However, gonadal hormonal analyses were outside the scope of the present thesis.

Procedures were standardized across participants and timepoints. Approximately one week before each clinic visit, participants received a secure Castor EDC link to complete the questionnaire battery. The baseline battery took around 60–70 minutes, whereas the shorter 3-month follow-up battery took considerably less time (i.e., approximately 40–50 minutes). Questionnaires were completed before, or at the latest on the day of, the behavioral assessment. Participants received financial reimbursement structured as €20 for the baseline questionnaire battery, €14 for the follow-up questionnaires, and €11 for each in-person behavioral session. Additionally, a certificate of participation was provided following the completion of each measurement timepoint.

Behavioral testing was conducted during a 60-minute in-person session on a dedicated laboratory laptop. Participants completed five behavioral tasks in total. The social network mapping task in Network Canvas was administered first, followed by four PsychoPy tasks implemented in PsychoPy v2024.2.4 (Peirce et al., 2019), which were self-programmed during the Capita Selecta project (i.e., Radboud University's Research Master Cognitive Neuroscience course). The Face-Word Emotional Stroop Task was consistently administered as the third task and took approximately 8 minutes to complete.

Power Analysis Details

Figure S3 presents the sensitivity analysis used to contextualize the initially anticipated sample size. The calculation was performed in R using the *pwr* package for paired *t*-test sensitivity estimation (Champely et al., 2020). For each sample size, pwr::pwr.t.test() was used to estimate the standardized effect required for 80% power at $\alpha = .05$, using a onesided paired test for the directional H1 main effect and a two-sided paired test for H2–H4. A ROPE buffer of ±0.1 SES was then added so that the minimum detectable effect (MDE) would exceed the preregistered negligible-effect interval of [−0.1, 0.1].

This paired-contrast approximation yielded MDE d_z = 0.56 for H1 and d_z = 0.63 for non-directional interaction contrasts at N = 30. Translated into RT units using an assumed baseline RT SD = 61.4 ms (Agustí et al., 2017), and the formula MDE = SES × SD × $\sqrt{[2(1 - r)]}$, these thresholds correspond to approximately 35–49 ms for H1 and 39–55 ms for H2–H4 across assumed test-retest correlations of r = 0.5 to r = 0.0. The identical curve for two-way (H2–H3) and three-way (H4) interactions reflects the simplified paired-contrast approximation, which does not equal to sensitivity in the final Bayesian mixed-effects model. In practice, higher-order interactions are typically estimated with lower precision because they depend on additional cell contrasts, multiple variance components, and both participant- and stimulus-level uncertainty.

Randomization Error

Between May 4 and October 2, predominantly during baseline assessments, the Face-Word Emotional Stroop Task was administered with an earlier PsychoPy script in which the intended stimulus-level randomization was not fully implemented. The task still presented the planned trial types, including happy and angry faces, happy and angry words, congruent and incongruent trials, and face-target and word-target blocks. However, the old script assigned each selected facial stimulus to one congruency condition in the first block and the opposite congruency condition in the second block, instead of ensuring that the same stimulus was properly balanced across congruency within each target-modality block. As a result, the overall task structure remained usable, but some facial stimuli were not evenly represented across all intended condition combinations. This produced a stimulus-level imbalance in part of the dataset, which was corrected in the updated task script used from October 2 onward.

Modelling Contrasts

Congruency was coded as congruent = −0.5 and incongruent = +0.5; timepoint as baseline = −0.5 and three months = +0.5; distractor valence as happy = −0.5 and angry = +0.5; block type as word-target = −0.5 and face-target = +0.5; and stimulus sex as male = −0.5 and female = +0.5.

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